

24UAM511 | NATURAL LANGUAGE PROCESSING

Course Syllabus · AI & ML Engineering · III Year

Nature of Course	Laboratory Integrated Theory (LIT) · PCC	L-T-P-C	2-0-2-3
Prerequisites	Python Programming, Machine Learning	Total Hours	60 Hrs

Course Objectives

Enable the students to:

1	Learn the fundamentals of Natural Language Processing (NLP) — including its origin, challenges, linguistic foundations, and core text processing techniques.
2	Understand statistical models at the word level — including N-gram models and Part-of-Speech tagging using machine learning approaches.
3	Equip students with techniques for syntactic analysis using context-free grammars, parsing algorithms, and probabilistic parsing methods.
4	Explore the representation of meaning in language through semantic and pragmatic analysis, including word sense disambiguation and logic-based methods.
5	Familiarize with discourse analysis, reference resolution, NLP lexical resources, and real-world applications using ML and DL models.

Course Contents

UNIT I INTRODUCTION TO NLP	CO1	6 Hrs
Origins and challenges of NLP — Language Modeling: Grammar-based LM, Statistical LM — Regular Expressions — Finite-State Automata — English Morphology — Transducers for lexicon and rules — Tokenization — Detecting and Correcting Spelling Errors — Linguistic Basics — Minimum Edit Distance.		
UNIT II WORD LEVEL ANALYSIS	CO2	6 Hrs
Unsmoothed n-grams — Evaluating n-grams — Smoothing — Interpolation and Backoff — Word Classes — Part-of-Speech Tagging: Rule-based, Stochastic, and Transformation-based tagging — Issues in PoS tagging — Hidden Markov and Maximum Entropy models.		
UNIT III SYNTACTIC ANALYSIS	CO3	6 Hrs
Context-Free Grammars — Grammar rules for English — Treebanks — Normal Forms for grammar — Dependency Grammar — Syntactic Parsing — Ambiguity — Dynamic Programming parsing — Shallow parsing — Probabilistic CFG — Probabilistic CYK — Probabilistic Lexicalized CFGs — Feature structures — Unification of feature structures.		

UNIT IV SEMANTICS AND PRAGMATICS	CO4	6 Hrs
Requirements for representation — First-Order Logic — Description Logics — Syntax-Driven Semantic analysis — Semantic attachments — Word Senses — Relations between Senses — Thematic Roles — Selectional restrictions — Word Sense Disambiguation: WSD using Supervised, Dictionary & Thesaurus, Bootstrapping methods — Word Similarity using Thesaurus and Distributional methods.		

UNIT V DISCOURSE ANALYSIS AND LEXICAL RESOURCES	CO5	6 Hrs
Discourse segmentation — Coherence — Reference Phenomena — Anaphora Resolution using Hobbs and Centering Algorithm — Coreference Resolution — Resources: Porter Stemmer, Lemmatizer, Penn Treebank, Brill's Tagger, WordNet, PropBank, FrameNet — Machine Learning and Deep Learning Approaches in NLP: LSTM — Applications of NLP: Sentiment Analysis, Machine Translation, Chatbots.		

Practical Exercises (30 Hrs)

#	Experiment	CO
1	Perform word-level analysis on a user-input text using Natural Language Processing.	CO1
2	Apply N-Gram language modeling techniques for user-input text.	CO2
3	Perform Part-of-Speech (POS) tagging on user-input text using NLTK's default POS tagger.	CO2
4	Implement chunking by defining grammar rules for extracting meaningful phrases.	CO3
5	Implement Word Sense Disambiguation using the simplified algorithm.	CO4
6	Implement sentiment analysis by using pre-trained models.	CO5
7	Implement a rule-based chatbot that responds with predefined answers to specific user inputs.	CO5
8	Open Ended Experiment.	CO1-CO5

Course Outcomes

At the end of the course, the student will be able to:

CO1	Understand the fundamentals of NLP — linguistic basics and basic processing techniques.
CO2	Analyze and apply statistical models at word level, including N-grams and PoS tagging.
CO3	Understand and apply syntactic analysis techniques including CFGs and probabilistic parsing.
CO4	Apply semantic and pragmatic methods such as FOL, WSD, and word similarity metrics.
CO5	Demonstrate discourse analysis and use NLP tools/resources for real-world applications.

Text Books

1	Raymond S. T. Lee, <i>Natural Language Processing: A Textbook with Python Implementation</i> , 2nd Edition, Springer Nature, 2025.
2	Daniel Jurafsky and James H. Martin, <i>Speech and Language Processing</i> , 3rd Edition, Prentice Hall, 2024.

Reference Books

1	Mohamed Zakaria Kurdi, <i>Natural Language Processing and Computational Linguistics 2: Semantics, Discourse and Applications</i> (Cognitive Science), 1st Edition, Wiley-ISTE, 2024.
2	Breck Baldwin, <i>Language Processing with Java and LingPipe Cookbook</i> , Kindle Edition, Packt Publishing, 2015.
3	Christopher D. Manning and Hinrich Schütze, <i>Foundations of Statistical Natural Language Processing</i> , MIT Press, 2018.
4	Steven Bird, Ewan Klein and Edward Loper, <i>Natural Language Processing with Python</i> , 1st Edition, O'Reilly Media, 2009.